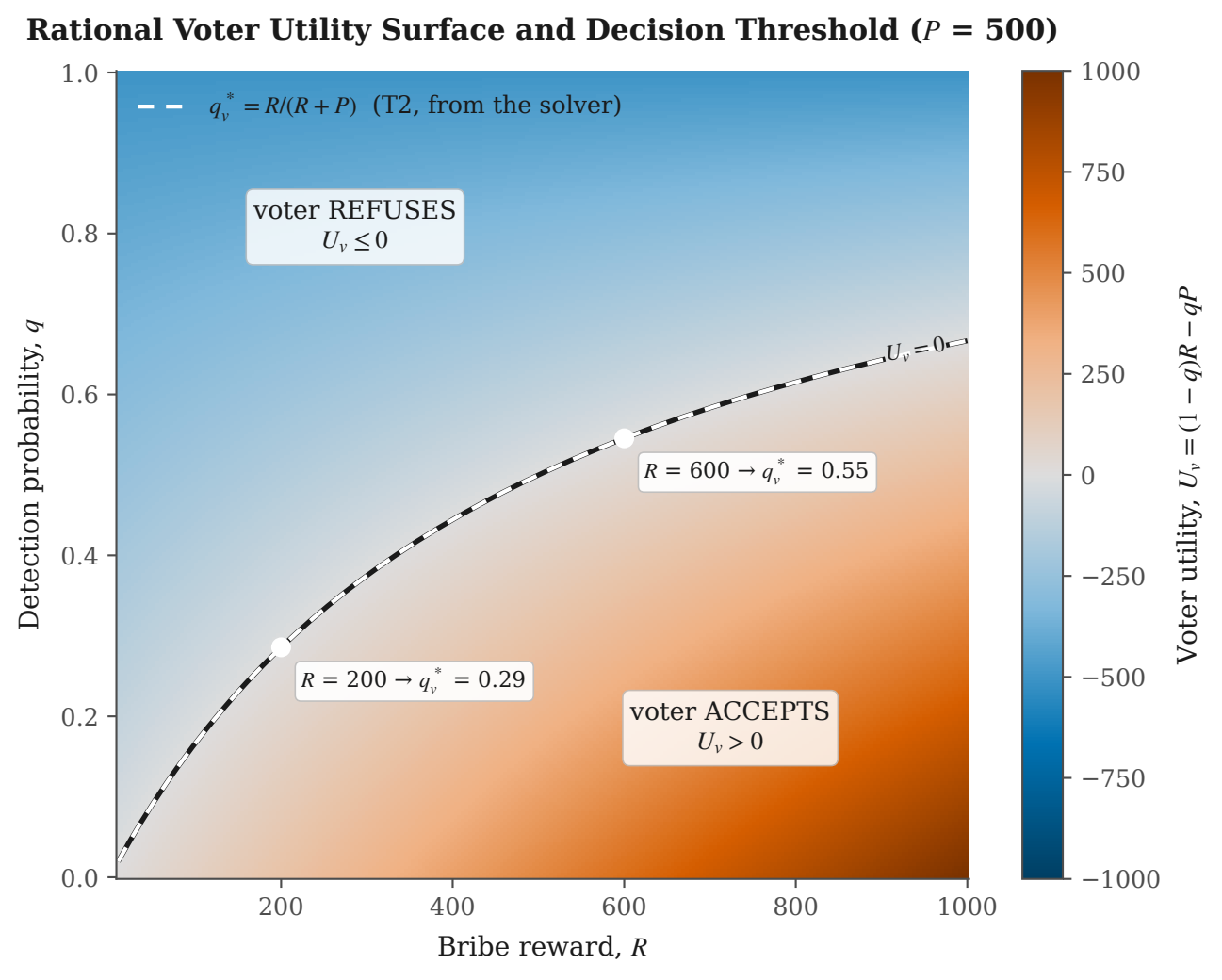

Rational Voter Utility Surface and Decision Threshold ($P$ = 500)
$q_v^* = R/(R+P)$ (T2, from the solver)
voter REFUSES
$U_v \le 0$
voter ACCEPTS
$U_v > 0$
$U_v = 0$
$R$ = 600 → $q_v^*$ = 0.55
$R$ = 200 → $q_v^*$ = 0.29
Detection probability, $q$
Bribe reward, $R$
Voter utility, $U_v = (1-q)R - qP$
0.0
0.2
0.4
0.6
0.8
1.0
200
400
600
800
1000
1000
750
500
250
0
−250
−500
−750
−1000